ABHAY DESHMUKH

VLSI AND RTL DESIGN CANDIDATE

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PROFESSIONAL SUMMARY

Final-year Electronics and Communication Engineering student (CGPA 8.7; expected graduation April 2027) building a foundation in digital RTL design using Verilog HDL. Developed educational RTL projects covering combinational logic, sequential buffering, and serial transmission, with testbenches and waveform-dump support. Seeking entry-level Graduate Engineer, RTL Design, or Digital Verification opportunities.

TECHNICAL SKILLS

HDL and RTL: Verilog HDL, combinational and sequential logic, opcode selection, finite-state machines, counters, pointers, parameterized modules
Verification: Directed testbench checks, reset and status-flag checks, serial-frame checks, VCD waveform generation
Programming: C, Embedded C, Python
Related systems: ESP32, sensor interfacing, MQTT, AWS IoT Core, communication-signal analysis with NumPy and TensorFlow-Keras

PROJECTS

4-bit ALU using Verilog HDL | GitHub repository
- Designed a combinational 4-bit ALU supporting eight opcode-selected arithmetic, logical, comparison, and pass-through operations, plus a zero-result flag.
- Added a directed self-checking testbench and VCD output for waveform inspection.

Focus: Verilog HDL, combinational RTL, case-based control, testbench

Synchronous FIFO using Verilog HDL | GitHub repository
- Created a parameterized single-clock FIFO with configurable data width and depth, read/write pointers, occupancy count, active-low asynchronous reset, and full/empty flags.
- Included testbench scenarios for reset, FIFO data ordering, and empty-state behavior, with VCD waveform output.

Focus: Verilog HDL, sequential logic, pointers and counters, parameterized RTL

UART Transmitter using Verilog HDL | GitHub repository
- Implemented an FSM-based UART transmitter for 8-N-1 framing, including a start bit, eight least-significant-bit-first data bits, stop bit, configurable bit timing, and tx_busy status.
- Included a testbench for transmitting byte 0xA5 and generating VCD waveforms.

Focus: Verilog HDL, FSM, counters, serial communication, testbench

EXPERIENCE AND ADDITIONAL PROJECT

Embedded Systems Intern — Pantech Prolabs India | 20 February – 20 May
- Hands-on training in Embedded C, microcontroller peripheral interfacing, and firmware development.

IoT-Enabled Predictive Maintenance System | GitHub profile
- Built an ESP32 multi-sensor monitoring node and streamed data to AWS IoT Core using MQTT; explored ANN and ensemble models for fault classification.

EDUCATION AND CERTIFICATIONS

B.Tech, Electronics and Communication Engineering — TGPCET, Nagpur | CGPA: 8.7 | Expected April 2027
HSC: 83.60% | SSC: 89.60% | Jawahar Navodaya Vidyalaya, Yavatmal | Certifications: AWS, Python, IoT, MATLAB